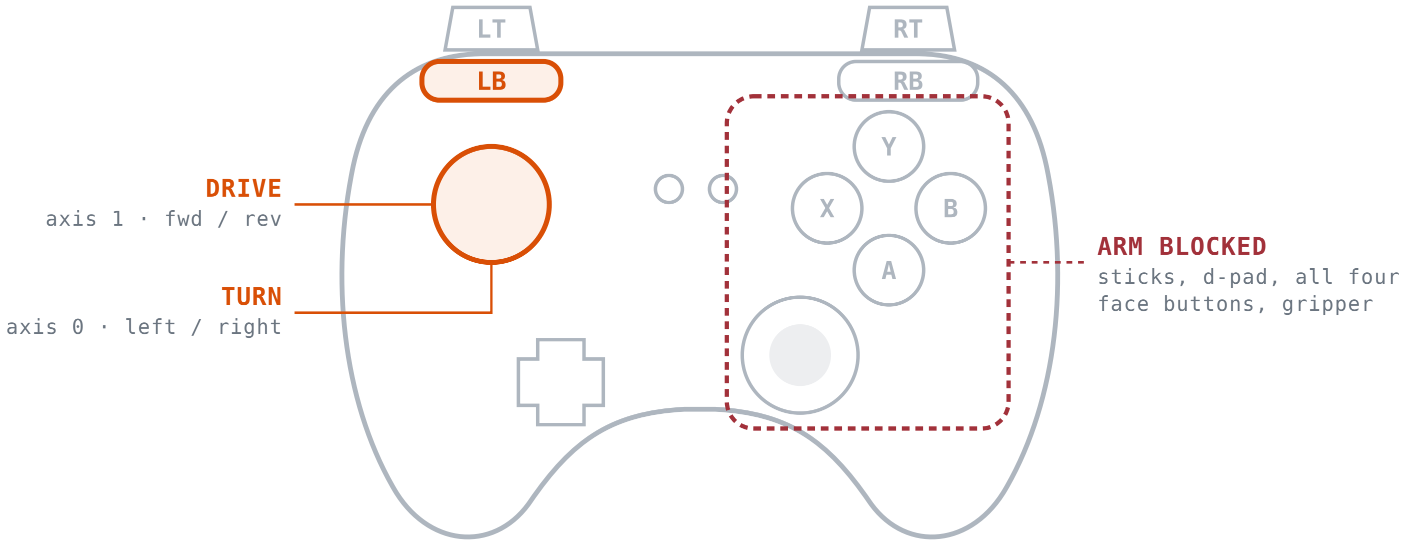
LB Rover drive

MOCK_ROVER_DRIVE · ROVER_CMD_VEL_JOYSTICK



Only the left stick is live. Holding LB hard-stops the arm before anything else is read.

LIVE CONTROLS

L stick ▲▼	**Drive forward / reverse** axis 1 · 0.55 m/s mock, 0.45 m/s ODrive
L stick ◀▶	**Turn** axis 0 · 1.60 rad/s mock, 1.40 rad/s ODrive

INERT WHILE LB IS HELD

Everything	Right stick, d-pad, A / B / X / Y and the whole arm do nothing.

The arm is actively blocked, not just idle. The teleop node checks LB first, commands a hard zero stop, and returns before reading any other control — it logs `LB rover mode active: arm blocked`. Stick deadzone here is 0.08, wider than the arm's 0.04.

NO MODIFIER
Nothing moves. Teleop publishes **silence**, not zeros.

BOTH RB + RT
RB wins. A brushed trigger can't change the sticks.

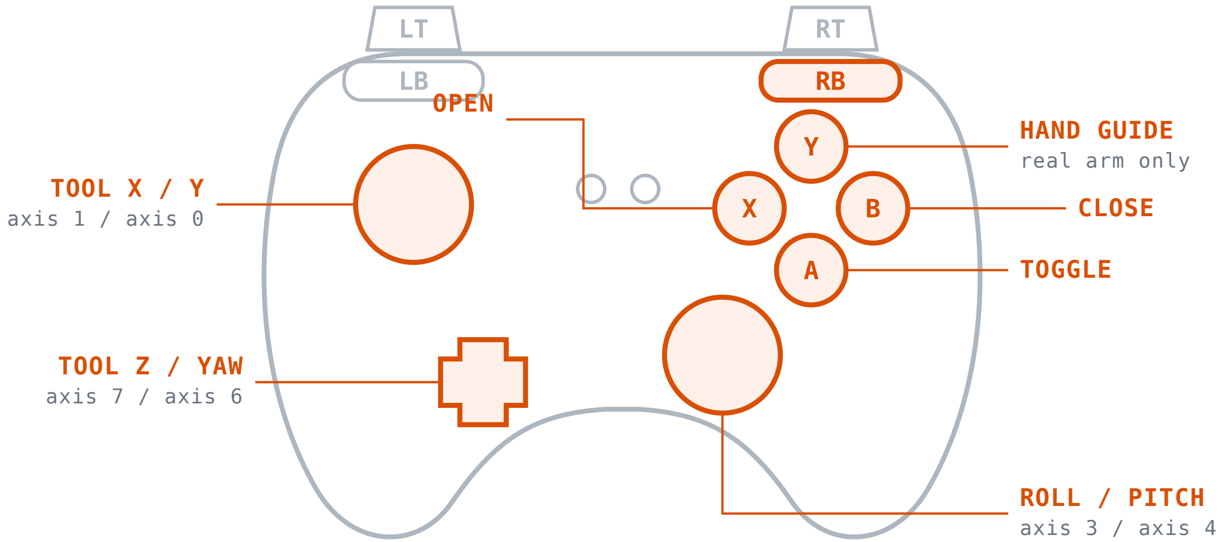
TRIGGER THRESHOLD
LT / RT engage past **0.50**, release below **0.35**.

SIGNAL LOSS
`/joy` silent **0.35 s** → arm stops.

RB

Arm — tool space

CARTESIAN IN BASE_LINK · REBEL_SERVO_TELEOP_GAMEPAD



Translation on the left stick, rotation on the right, Z and yaw on the d-pad. The whole face cluster is gripper work, except Y.

TRANSLATE — 0.10 M/S AT FULL DEFLECTION

L stick ▲▼	**Tool X** axis 1
L stick ◀▶	**Tool Y** axis 0
D-pad ▲▼	**Tool Z** axis 7

ROTATE — 0.18 RAD/S, ABOUT 10°/S

R stick ◀▶	**Roll** axis 3
R stick ▲▼	**Pitch** axis 4
D-pad ◀▶	**Yaw** axis 6

GRIPPER & HAND GUIDING

X · hold	**Open** 2.0 rad/s, full stroke ~0.82 s
B · hold	**Close**
A · press	**Toggle** open ↔ closed
Y · hold	**Hand guiding** — ZeroTorque, real arm only

Release overshoot is real. The arm keeps travelling roughly 0.25 s after you let go — transport delay plus the ReBeL's own firmware deceleration, neither reachable from ROS. Release slower to stop shorter.

NO MODIFIER
Nothing moves. Teleop publishes **silence**, not zeros.

BOTH RB + RT
RB wins. A brushed trigger can't change the sticks.

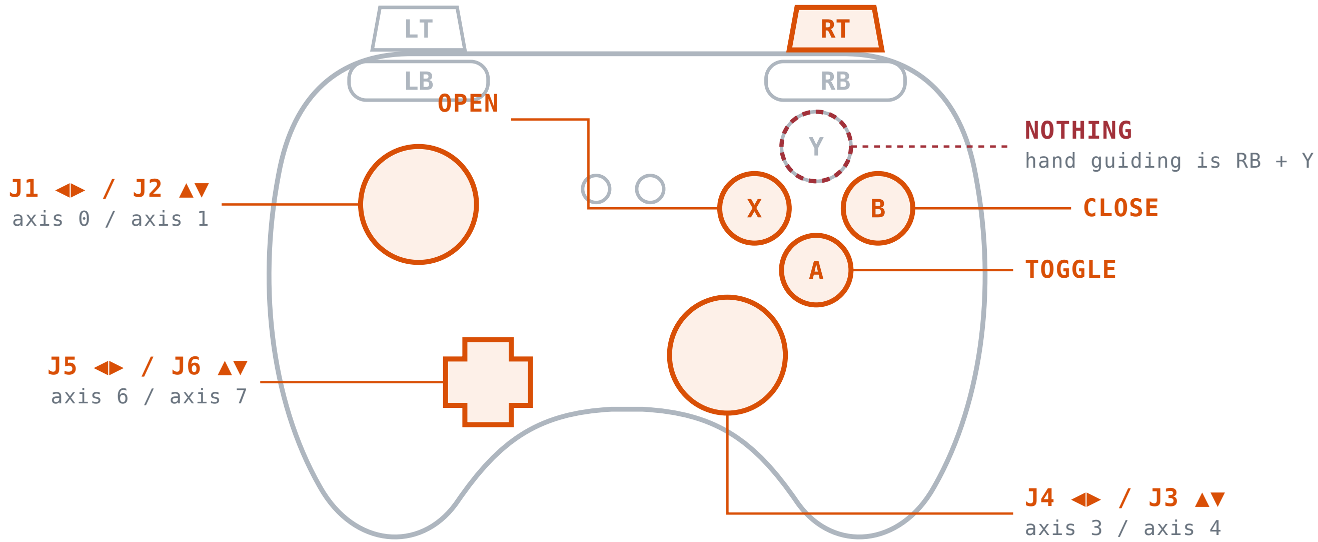
TRIGGER THRESHOLD
LT / RT engage past **0.50**, release below **0.35**.

SIGNAL LOSS
/joy silent **0.35 s** → arm stops.

RT

Arm — joint jog

ONE JOINT PER STICK AXIS · REBEL_SERVO_TELEOP_GAMEPAD



Same sticks, different meaning: six joints instead of a tool frame. Y goes dead here — hand guiding needs RB.

STICKS — 0.38 RAD/S PER JOINT

L stick ◀▶	**joint1** axis 0 · base yaw
L stick ▲▼	**joint2** axis 1 · shoulder
R stick ▲▼	**joint3** axis 4 · elbow
R stick ◀▶	**joint4** axis 3 · forearm roll

D-PAD — THE WRIST

D-pad ◀▶	**joint5** axis 6 · wrist pitch
D-pad ▲▼	**joint6** axis 7 · wrist roll

GRIPPER — IDENTICAL TO RB

X · hold	Open
B · hold	Close
A · press	Toggle
Y	Nothing. Hand guiding is RB + Y, never RT + Y.

0.28 rad/s is a ceiling, not the speed. If any joint exceeds it the whole velocity vector scales down uniformly, so direction is preserved — but near a singularity the tool moves visibly slower than the number suggests.

NO MODIFIER
Nothing moves. Teleop publishes **silence**, not zeros.

BOTH RB + RT
RB wins. A brushed trigger can't change the sticks.

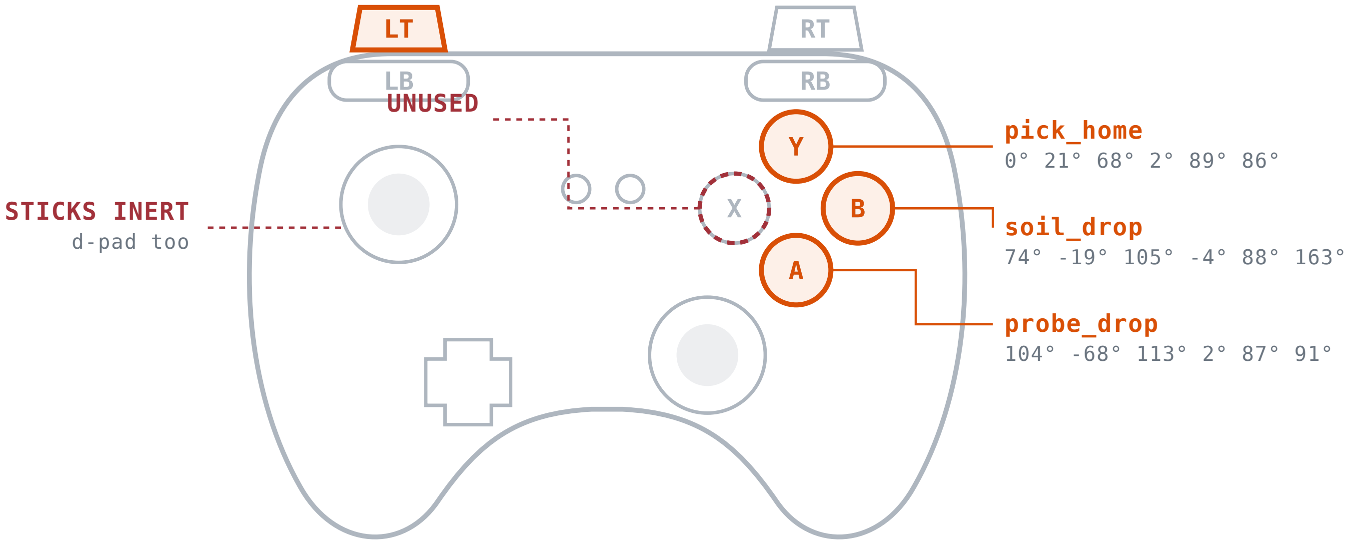
TRIGGER THRESHOLD
LT / RT engage past **0.50**, release below **0.35**.

SIGNAL LOSS
/joy silent **0.35 s** → arm stops.

LT

Arm — stored poses

PLANNED MOVEGROUP MOVES · ARM_PRESET_POSE_JOYSTICK



Three stored poses on the face buttons. Each is a planned move, collision-checked the same way RViz Plan & Execute is.

STORED POSES — PRESS, DON'T HOLD

Y	**pick_home** `0° 21° 68° 2° 89° 86°`
A	**probe_drop** `104° -68° 113° 2° 87° 91°`
B	**soil_drop** `74° -19° 105° -4° 88° 163°`
X	Unassigned.
Sticks	Inert. Presets are the only thing LT does.

REFUSED — LOGGED, NEVER SILENT

RB held	That combo is hand guiding.
RT held	Arm is under joint jog.
LB held	Rover has the sticks.
Mid-move	A preset is already running.
Two at once	Refuses rather than picking one arbitrarily.

> **The gripper is left alone.** None of the three poses commands the jaw, so a preset can be used while holding a probe. Moves run at 0.25 velocity and acceleration scale. Watch `/arm_preset_pose/status`.

CANONICAL LAYOUT
As `joy_layout_normalizer` republishes it — driver-independent.

TRIGGERS ARE AXES
LT = axis 2, RT = axis 5, **0.0** released to **1.0** pressed.

MAPPING & SPEEDS
`gamepad.yaml`, `teleop_speeds.yaml`

STORED POSES
`aries.srdf`